Workflow for Custom NER with DistilBERT and Active Learning

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1. Data Preparation

1.1. Clean and Preprocess Data

- **Columns:**
  - **Question:** Original question text.
  - **Answer:** Original answer text.
  - **Unique_Keywords:** A tokenized list generated by your custom tokenizer (removes single-letter words, numbers, punctuation; converts to lower case; removes stop words; lemmatizes/stems).
- **Goal:**
  - Clean the raw text in the question and answer columns to make it friendly for the DistilBERT tokenizer.
  - Use the custom tokenizer output for domain-specific keyword extraction.

1.2. Generate Domain-Specific Entity Definitions

- **Define Custom NER Labels:**
  - Examples: CHARACTER, CREATURE, LOCATION, SPELL, HOGWARTS_HOUSE, QUIDDITCH, etc.
- **Supplement with External Sources:**
  - Scrape or collect comprehensive lists from reliable web sources to ensure your custom labels cover all relevant entities.
- **Purpose:**
  - These lists guide your annotation process and serve as reference points to ensure consistent labeling.

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2. Annotation & Active Learning Setup

2.1. Initial Annotation

- **Select a Small Subset:**
  - Manually annotate around 150–200 Q&A pairs covering a diverse range of entity types.
- **Annotation Tool:**
  - Use an annotator tool (e.g., Doccano, Prodi.gy, or Brat) to label tokens in both questions and answers with your custom entity labels.
- **Output Format:**
  - Export the annotated data in a format compatible with your training framework (e.g., JSONL, CSV, or spaCy's format).

2.2. **Active Learning Integration**

- **Train an Initial Model:**
  - Fine-tune a pre-trained DistilBERT model on the initial annotated subset.
- **Pre-Label Unannotated Data:**
  - Use the trained model to predict NER labels on a new batch of unannotated Q&A pairs.
- **Human-in-the-Loop:**
  - Manually review and correct the model's predictions.
  - Add the corrected data to your training set.
- **Iterate:**
  - Retrain the model with the expanded training set.
  - Repeat the process (pre-label, correct, retrain) until the model reaches acceptable performance.

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3. Model Fine-Tuning

3.1. **Preprocess Final Inputs**

- **Dual Tokenization:**
  - Use your custom tokenizer for initial cleaning and keyword extraction.
  - Then, re-tokenize the cleaned text with the DistilBERT tokenizer to produce tokens in the expected format.
- **Data Formatting:**
  - Ensure the training data includes token IDs, attention masks, and corresponding token-level labels.

3.2. **Model Setup**

- **Load Pre-trained DistilBERT Model:**
  - Adjust the classification head to match your custom NER labels (set `num_labels` equal to the number of custom entity types).
- **Define Training Arguments:**
  - Specify hyperparameters (learning rate, batch size, number of epochs, etc.).

3.3. **Fine-Tune with Active Learning Data**

- Fine-tune the DistilBERT model on your training set (initial annotated data plus corrections from active learning iterations).
- Optionally apply data augmentation techniques (e.g., synonym replacement, back-translation) to enrich the training set.

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4. Evaluation and Iterative Improvement

4.1. Model Evaluation

- **Validation Set:**
  - Use a separate portion of your data for validation.
- **Metrics:**
  - Monitor precision, recall, and F1-score for each entity type.

4.2. Iterative Refinement via Active Learning

- **Error Analysis:**
  - Identify common misclassifications and annotate additional examples focusing on these errors.
- **Retraining:**
  - Update the training set with newly corrected annotations and retrain the model until performance stabilizes.

4.3. Final Testing

- Evaluate the final model on a holdout test set to ensure generalization.

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5. Post-Training and Deployment

5.1. Save the Fine-Tuned Model and Tokenizer

- Save the model and tokenizer locally or deploy them in your desired environment.

5.2. Inference Pipeline

- Set up an inference pipeline that:
  - Uses custom preprocessing for keyword extraction.
  - Tokenizes with the DistilBERT tokenizer.
  - Runs the NER model to label new Q&A pairs.
  - Optionally cross-checks against your comprehensive keyword lists for consistency.

Summary:

- **Preprocess** your data using your custom tokenizer for domain-specific cleaning and keyword extraction.
- **Manually annotate** a small, diverse set (150–200 Q&A pairs) for initial training.

- **Implement active learning:** Train an initial model, pre-label unannotated data, correct predictions, and iteratively expand your training set.
- **Fine-tune DistilBERT** with the updated training data and proper tokenization.
- **Evaluate and iterate** until satisfactory performance is achieved.
- **Deploy** the final model in an inference pipeline that leverages both your custom preprocessing and DistilBERT's contextual power.